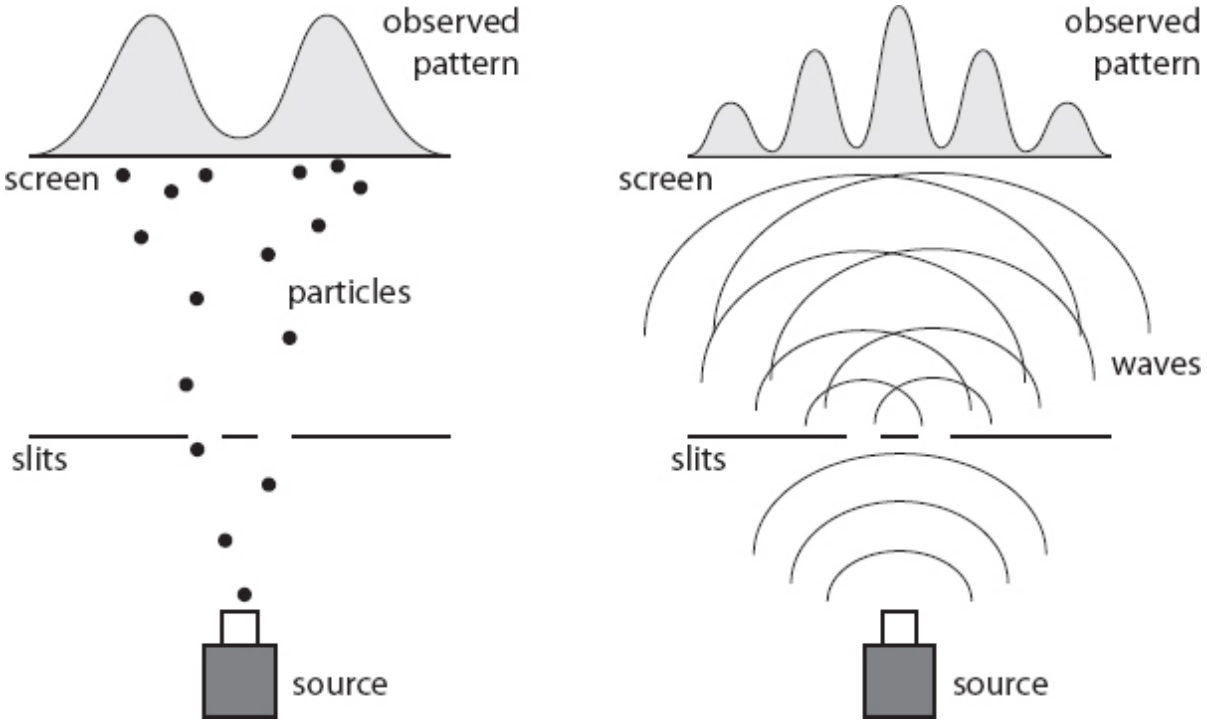


Now let's introduce two slits. (The slits have to be very close together for this to work, which is one reason it took so long for the experiment to actually be carried out.) Once again, electrons pass through the slits and leave individual marks on the screen on the other side. However, their marks do *not* clump into two lines, as the classical pellets did. Rather, they form a series of lines: a high-density one in the middle, surrounded by parallel lines with gradually fewer marks, each separated by dark regions with almost no marks at all.

In other words, electrons going through two slits leave what is unmistakably an interference pattern, just like waves do, even though they hit the screen with individual marks just like particles. This phenomenon has launched a thousand unhelpful discussions about whether electrons are "really" particles or waves, or are sometimes particle-like and other times wave-like. One way or another, it's indisputable that something went through both slits as the electrons traveled to the screen.

At this point this is no surprise to us. The electrons passing through the slits are described by a wave function, which just like our classical wave will go through both slits and oscillate up and down, and therefore it makes sense that we see interference patterns. Then when they hit the screen they are being observed, and it's at that point they appear to us as particles.